

Project Proposal: Developing A Standard Evaluation Framework for Event-Centric Knowledge Graphs

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ABSTRACT

Event knowledge graphs (EKGs) are critical for representing complex event-related data and capturing temporal and causal relationships among events and entities. High-quality evaluations of EKGs are crucial for ensuring their effectiveness and reliability in real-world applications. EKG's are designed to capture and represent the complex relationships and sequences of events, which are vital for tasks like event prediction, and causal analysis. Due to this, a robust evaluation framework is needed assess how accurately an EKG models these events, their temporal dynamics, and interdependencies. Without rigorous evaluation, the EKG may provide misleading insights, potentially leading to faulty decision-making in critical domains such as finance, healthcare, and cybersecurity. Despite the growing body of research and applications of EKGs, a significant gap remains in their evaluation. Current measures often address only a subset of necessary features, varying significantly between studies and projects. This which results in incomplete assessments that fail to encapsulate all critical aspects of EKGs.

My research aims to address this gap by developing a comprehensive evaluation framework specifically tailored for event knowledge graphs. This framework integrates existing evaluation methods while introducing new metrics to address gaps in traditional assessments, ensuring a comprehensive and simultaneous evaluation of the temporal, causal, contextual, and several other critical nuances inherent to EKGs. By systematically analyzing existing evaluation methods, identifying their limitations, and proposing a more suitable approach, this project seeks to enhance the accuracy and relevance of EKG evaluations. The proposed framework will facilitate more robust assessments, guiding future research and applications in the domain of event-centric knowledge representation

Keywords

Knowledge Graphs, Event-Centric Knowledge Graphs

1. RESEARCH DESCRIPTION

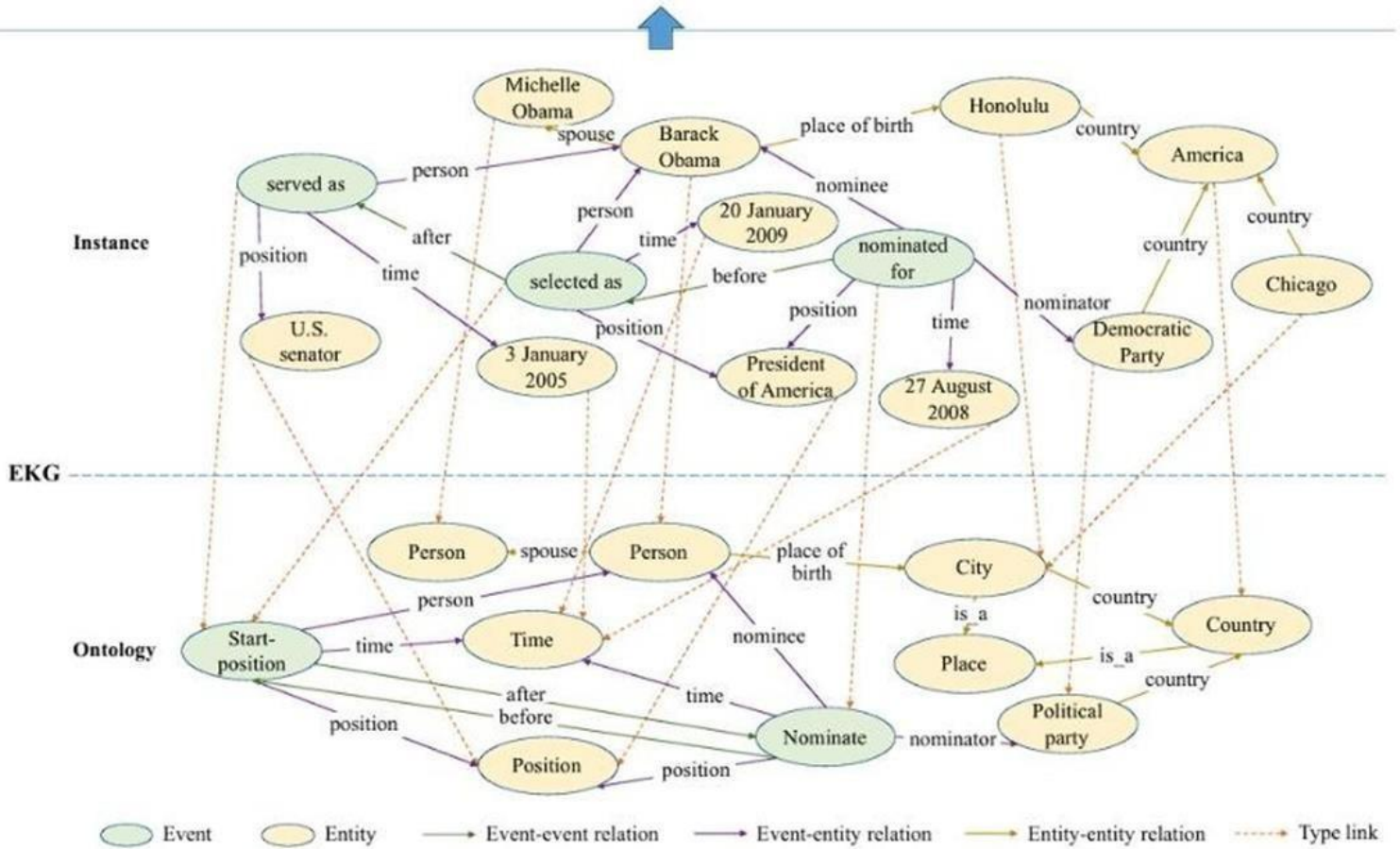
Event knowledge graphs (EKGs) are advanced data structures designed to represent and organize information around events rather than static entities.[1] Unlike traditional knowledge graphs, which focus primarily on entities and their relationships, EKGs emphasize the temporal, causal, and contextual connections between events and the entities involved. This makes EKGs particularly effective in domains where understanding the sequence, cause, and impact of events is crucial, such as in news analytics, historical data analysis, and event-driven systems. For instance, ECKGs constructed from news articles can capture dynamic changes over time, helping users reconstruct complex

storylines and explore hidden patterns that are often overlooked in traditional knowledge graphs [2].

Traditional Knowledge Graphs are designed to represent factual information about entities (such as people, places, organizations, etc.) and their relationships. The primary goal is to provide a comprehensive and structured representation of knowledge, often used for tasks like semantic search, question answering, and information retrieval. While EKG's focus specifically on representing events and the relationships between them. The aim is to capture the temporal, spatial, and causal relationships between events, providing a dynamic understanding of sequences and interactions that occur over time. [1] The two types of knowledge graphs differ in a myriad of ways, one of the most important is the way they represent relationships. Traditional KGs emphasize static relationships between entities, such as "Person A works at Company B" or "City X is in Country Y." The relationships are often more static and do not inherently incorporate time as a dimension. In contrast, EKG's emphasize dynamic relationships, including temporal (when events happen), causal (what causes what), and spatial (where events occur) relationships. For example, "Event A leads to Event B," "Event X occurs after Event Y," or "Event Z happens at Location L." [4].

EKGs solve several critical issues by capturing the dynamic nature of information. They provide a more nuanced understanding of how events unfold over time and how they influence each other, offering insights into causality and sequence that are often missed by entity-centric models. For example, in news analytics [2] EKGs can help trace the development of a story, identify pivotal moments, and understand the relationships between different news events and entities involved. This enables more accurate trend analysis, event prediction, and contextual understanding.

Traditional KGs emphasize static factual relationships (e.g., "Person A works at Company B"). They do not capture complex event relationships such as causality, temporality, or sequence of events (e.g., "Event X causes Event Y, which happens after Event Z"). This makes it difficult to use them for applications like decision-making or scenario planning that rely on understanding the progression and dependencies of events. To address this, EKGs go beyond simple entity-entity relationships by incorporating event-event relationships (such as causal, temporal, and conditional relationships) and event- entity relationships. This makes them more suitable for applications requiring deep reasoning about events, such as predicting outcomes, understanding causality, and generating timelines. [1]



Traditional KGs are also not well-suited to incorporate the context surrounding events, such as the circumstances under which an event occurred or its impact. This makes it challenging to perform more nuanced reasoning or to extract insights that depend on understanding the full context of an event sequence. EKG's, however, provide a framework to represent the context and attributes associated with each event, such as participants, locations, and time frames. This enhanced contextual understanding allows for more accurate and nuanced reasoning, improving the quality of insights that can be extracted. [4]

The current evaluation methods for Event Knowledge Graphs tend to focus on assessing specific features, such as accuracy of event representation, temporal relations, or completeness of spatio-temporal queries, in isolation rather than providing a comprehensive evaluation of all relevant aspects. Metrics like Precision and Recall are often used to evaluate the accuracy of event relationships by comparing system-generated graphs with manually annotated ones, while other methods, like the use of Cohen's Kappa Coefficient or Specific Triple Accuracy, assess the consistency and semantic richness of event references in news data [2]. Some approaches, like the ones used to evaluate AugGKG[5], evaluate completeness and efficiency in handling complex queries within geographic contexts. However, these methods usually address one or two dimensions—such as

event sequencing, causal relationship extraction, or multilingual support—without encompassing the full set of features that define an ECKG. Thus, while existing evaluation techniques provide valuable insights, there is a need for more holistic approaches that consider the diverse and interconnected attributes of ECKGs to ensure a balanced and thorough assessment of their capabilities.

Given these limitations, there is a clear need for a novel evaluation framework tailored specifically for ECKGs that can comprehensively assess their multifaceted nature. A robust evaluation framework must go beyond traditional metrics to incorporate new measures that capture the graph's ability to handle dynamic event evolution, support inferential reasoning, maintain consistency across diverse data sources, and represent events in a way that is both semantically rich and contextually accurate. Such a framework would enable a more nuanced understanding of an ECKG's strengths and weaknesses, guiding improvements in their construction and application. It would also provide a standardized way to compare different models, fostering advancements in ECKG research and development by highlighting which features are being successfully addressed and which require further innovation.

Ultimately, a comprehensive evaluation framework would ensure that ECKGs are fully capable of supporting the complex decision-making and analytical needs of real-world applications.

Thus, the primary objective of this research is to develop a comprehensive evaluation framework for event knowledge

graphs. This framework aims to integrate traditional EKG metrics with novel metrics and methodologies specifically designed to assess the temporal sequences, causal links, and contextual information that characterize EKGs. By incorporating these specialized metrics, the proposed framework will enable a more robust and meaningful evaluation process.

The research methodology begins with a systematic analysis of existing evaluation techniques for EKGs, involving a detailed review to understand their strengths and limitations. This critical examination will identify specific gaps in the current methods, particularly their shortcomings in effectively addressing the unique temporal and causal relationships within EKGs. Based on these insights, the research will then focus on developing a new evaluation framework tailored to the unique requirements of EKGs.

This new framework is expected to enhance the accuracy and relevance of EKG evaluations, ensuring a more precise identification of their strengths and weaknesses. Such improvements will lead to better-informed decisions in the development and application of EKGs. Furthermore, this framework is intended to serve as a benchmark for future research and practical applications in event-centric knowledge representation. By filling existing gaps and providing a robust evaluation method, it aims to set a new standard in the field, encouraging researchers to refine and build upon these evaluation practices.

2 PRIOR WORK

2.1 EKG Quality Evaluations

In prior research on evaluating EKGs, various approaches and metrics have been explored to measure their effectiveness in capturing complex event relationships. One notable study, "Discovering Event Evolution Graphs from News Corpora," [6] employs Precision and Recall metrics to assess the accuracy of event representations within EKGs. Here, nodes represent individual news events, and edges depict evolutionary relationships between them. The study evaluates these relationships by comparing an automatically generated graph to a manually created true structure, where precision is calculated as the ratio of correctly identified event relationships to the total identified by the system, and recall is the ratio of correctly identified relationships to the total number of correct relationships in manual annotations. This methodology effectively measures the system's ability to accurately identify relevant event relationships while assuming a consistent set of predefined events and focusing on differences in relationships.

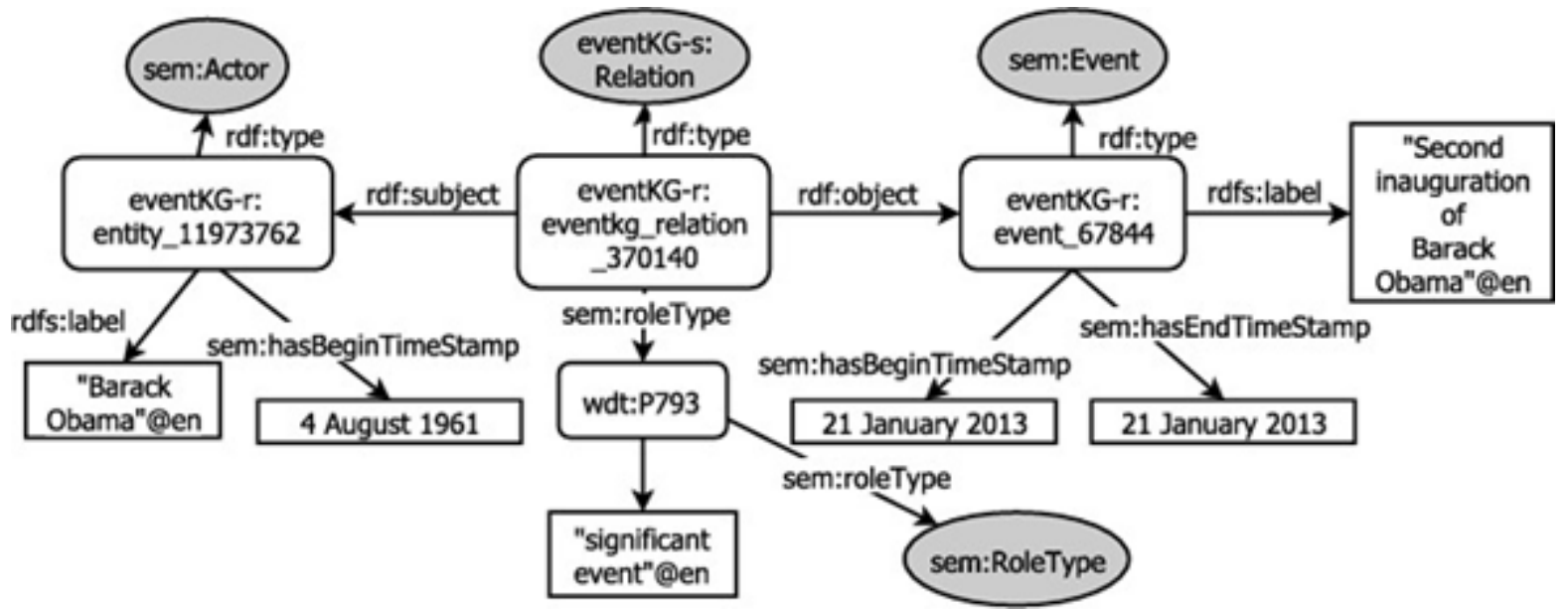
In contrast, other studies like "Construction of Event Knowledge Graph Based on Semantic Analysis by Yixin Song" [7] emphasize the application of semantic analysis and rule-based causality extraction methods in constructing EKGs focused on corporate credit events, but do not specify formal evaluation metrics. This indicates a more exploratory approach without direct comparison to a ground truth dataset, revealing a gap in standardized evaluation practices. Meanwhile, research on "Building Event-Centric Knowledge Graphs from News" adopts a more comprehensive evaluation framework, incorporating metrics such as Overall Accuracy, Specific Triple Accuracy, and Cohen's Kappa Coefficient.[2] These metrics provide a nuanced assessment of event representations by focusing on accuracy,

semantic richness, and the reliability of inter-rater evaluations. The use of multiple metrics in this study highlights the need for robust evaluation methods that capture the temporal and causal complexities unique to EKGs, underscoring the limitations of traditional metrics used in non-event-centric knowledge graphs. This varied landscape of approaches demonstrates the ongoing evolution and need for specialized evaluation frameworks tailored to the unique characteristics of EKGs. While these evaluations provide valuable insights into specific aspects such as accuracy, semantic richness, and inter-rater reliability, they fall short of assessing all the features inherent in EKGs, such as temporal dynamics, causal relationships, and contextual integration. As a result, there remains a need for a more comprehensive evaluation framework that can holistically capture the complex, interconnected nature of event-centric knowledge representation.

Another notable approach is presented in the evaluation of AugGKG[5]: A Grid-Augmented Geographic Knowledge Graph Representation and Spatio- Temporal Query Model. This study focuses on assessing the completeness, efficiency, and uniformity of accuracy in representing geographic data that integrates semantic, spatial, and temporal information. AugGKG models events as nodes with attributes like type and impact, facilitating complex spatio-temporal reasoning through the use of time slices for dynamic tracking of changes. The evaluation measures completeness by analyzing the correctness and comprehensiveness of responses to spatio-temporal queries across different models (AugGKG, YAGO, GeoKG, and GEKG). Human experts scored each model based on its ability to handle simple, regular range, and complex deductive queries, with AugGKG excelling in managing implicit relationships that other models could not handle. The study also assesses efficiency through query performance tests, revealing that AugGKG's grid-based system allows for significantly faster query times. Uniformity of accuracy is measured by the model's consistency in handling heterogeneous spatio-temporal data, with AugGKG demonstrating superior adaptability in maintaining uniform accuracy across queries, thanks to its innovative grid-based encoding. While the evaluation of AugGKG effectively addresses several key features, such as spatio-temporal reasoning, completeness, and query performance, it falls short in comprehensively assessing all the critical features of EKGs, including causal relationship representation, reasoning and inference capabilities, flexibility and extensibility, which are essential for a holistic evaluation of EKGs.

Another significant contribution comes from the evaluation of the Chinese Travel Domain Event Evolutionary Graph (EEG)[8], which focuses on capturing evolutionary patterns and development logic between events in the travel domain. This graph uses a directed cyclic structure to represent sequential and causal relationships between generalized event phrases. The evaluation process involves training classifiers to identify these relationships and using metrics such as accuracy, precision, recall, and F1- score to assess their performance. These metrics evaluate the classifier's ability to correctly predict sequential relations and their directions, ensuring the model's effectiveness in understanding event evolution. The model's predictions are validated using human raters or a separate validation dataset to ensure reliability and applicability in real-world data.

Additionally, the Event Logic Graph (ELG) [9] study by Xiao Ding, Zhongyang Li, Ting Liu, and Kuo L explores the representation of events and their relationships in specific domains like finance and travel. The ELG uses directed edges for



sequential, causal, and conditional relationships and supports cyclic structures to capture repeated or influencing events. The study employs various metrics, including accuracy, F1 score, precision, and recall, to evaluate the effectiveness of identifying and classifying these relationships, as well as predicting event sequences. The evaluation leverages BERT-based approaches, rule-based methods, and different model combinations to assess relationship extraction and event prediction, providing a comprehensive understanding of event logic and its application.

Prior research on the evaluation of EKGs also introduced various metrics to assess their effectiveness in representing events across different languages, temporal spans, and domains. For example, EventKG:[3] A Multilingual Event-Centric Temporal Knowledge Graph employs several evaluation metrics, including completeness, coverage, consistency, and multilingual and temporal accuracy. The metrics are designed to measure aspects such as the number of events and relations, the coverage of temporal and spatial information, consistency in event fusion, and the quality of multilingual representation. For instance, completeness improvement is gauged by comparing coverage before and after integrating additional sources, while interlinking assesses the percentage of events linked to multiple reference sources, reflecting the graph's data integration quality. These diverse metrics offer a comprehensive framework for evaluating the breadth and depth of event representations, especially in a multilingual and temporally rich context.

Both the Event Logic Graph (ELG) study by Xiao Ding et al. and the EventKG paper fall short of fully evaluating all EKG features. While the ELG study focuses on relationship extraction and event prediction, it lacks evaluation on temporal reasoning, scalability, contextual completeness, and reasoning capabilities. Similarly, the EventKG paper covers aspects like completeness, coverage, and multilingual accuracy but does not address scalability, causal

reasoning, , or dynamic event updating. Both papers offer valuable insights but need a more comprehensive evaluation framework to fully capture the diverse and complex features essential for robust EKGs.

In contrast, other event-centric knowledge graphs like OEKG[10] do not specify particular evaluation metrics, highlighting a gap in standard evaluation practices. Meanwhile, ASER: A Large-scale Eventuality Knowledge Graph focuses on both intrinsic and extrinsic evaluations to assess the quality of eventualities and relation extraction, demonstrating high accuracy rates and strong performance in natural language understanding tasks. Intrinsic evaluation is based on human annotation accuracy for eventualities and relations, achieving high agreement rates, while extrinsic evaluation involves testing on real-world tasks like the Winograd Schema Challenge to validate its effectiveness in resolving pronoun references. These varied approaches underscore the evolving methodologies for EKG evaluation, emphasizing the need for robust, domain-specific frameworks that can capture the unique characteristics and complexities of event-centric knowledge graphs. Event-centric knowledge graphs like OEKG lack defined evaluation metrics, highlighting a need for standardized assessment practices. In contrast, ASER uses both intrinsic and extrinsic evaluations, demonstrating high accuracy and effectiveness in natural language understanding tasks, but still underscores the need for domain-specific evaluation frameworks for EKGs.

2.2 Knowledge Graph Evaluation frameworks

In the evaluation of traditional knowledge graphs, the KGrEaT[11] framework, proposed by Heist, Hertling, and Paulheim, introduces a novel approach to assess knowledge graph quality based on their utility in downstream tasks, such as classification, clustering, and recommendation. Traditional evaluations often focus on intrinsic metrics like accuracy and completeness, which, while valuable, do not fully capture the effectiveness of a knowledge graph in real-world applications. KGrEaT addresses this gap by providing a fixed evaluation setup where different knowledge graphs are compared using the same

tasks, allowing for a more comprehensive understanding of their utility. The framework automatically maps the input knowledge graph to relevant datasets and calculates performance metrics for predefined tasks, offering insights into how well different graphs perform in practical scenarios. KGrEaT’s modular design supports easy extension with new tasks and datasets, making it a flexible tool for the research community to evaluate and improve knowledge graphs. By focusing on extrinsic task-based metrics, KGrEaT enables more informed decisions when selecting or refining knowledge graphs for specific applications. While KGrEaT is effective for evaluating traditional knowledge graphs using downstream tasks, its expansion to Event Knowledge Graphs (EKGs) faces several limitations. EKGs capture complex, dynamic relationships between events involving temporal, spatial, and causal dimensions, which require evaluations beyond standard metrics like accuracy or F1-score. KGrEaT’s current tasks—designed for static, entity-based graphs—do not account for the temporal reasoning, causal inference, and event sequence prediction crucial for EKGs. Adapting KGrEaT for EKGs would need new tasks, specialized metrics, and advanced mapping techniques to handle the contextual nature of events. Furthermore, while KGrEaT’s modularity allows for some flexibility, incorporating EKG-specific preprocessing and evaluation setups would require significant customization, making it a challenging but necessary step to evaluate the unique properties of EKGs effectively.

Chen et al. [12] proposed a practical framework for evaluating the quality of knowledge graphs, focusing on key quality dimensions that are both feasible and relevant for real-world applications. Their framework emphasizes essential metrics such as syntactic validity, semantic accuracy, completeness, conciseness, and trustworthiness, tailored to assess whether a knowledge graph is "fit for purpose." These metrics are measured using targeted, scalable methods. For instance, syntactic validity is verified through schema-based validation tools to detect syntax errors, while semantic accuracy is assessed using machine learning models or manual reviews to identify and correct incorrect or misleading information. Completeness is measured by comparing the graph’s data against reference datasets using sampling techniques, and conciseness is evaluated by detecting redundant triples. The framework also includes trustworthiness, which assesses the credibility of data sources. By focusing on these core dimensions and employing practical evaluation methods like sampling-based approaches, Chen et al.’s framework provides a more streamlined and effective approach for ensuring the quality of large-scale knowledge graphs in knowledge discovery and application development contexts.

While Chen et al.’s framework is effective for general knowledge graphs, it might not be as suitable for evaluating EKGs due to the unique requirements and complexities associated with event data. EKGs inherently focus on dynamic and temporal information, involving not just entities and their relationships but also sequences of events, durations, causality, and temporal reasoning. Unlike static knowledge graphs, which emphasize syntactic validity, semantic accuracy, and data completeness in a more straightforward manner, EKGs require specialized metrics that account for the temporal ordering of events, the accuracy of event sequences, the reasoning capabilities to infer future events, and the representation of causal relationships between events.

For example, while Chen et al.’s framework assesses completeness by comparing data against reference datasets, this approach may not capture the dynamic nature and evolving contexts in EKGs, where new events can continuously alter the "completeness" of the graph. Similarly, while the framework’s focus on conciseness—such as detecting redundant triples—is beneficial for static data, it doesn’t directly address the need to evaluate the coherency and accuracy of temporal event sequences or the ability of an EKG to perform complex temporal and causal reasoning.

3. PROBLEM STATEMENT

As previously, despite their growing significance, the evaluation of EKGs remains a significant challenge. Current evaluation methods are often fragmented, addressing only a subset of the features that define an EKG. For instance, existing methods may focus on the accuracy of event representation or temporal relationships but fail to provide a comprehensive assessment that considers all the critical dimensions of an EKG, such as causal links, contextual information, and the ability to handle complex queries involving multiple events. This lack of a standardized and comprehensive evaluation framework results in incomplete assessments that do not fully capture the capabilities and limitations of EKGs. Consequently, EKGs may fall short in practical applications, potentially leading to misleading insights, incorrect predictions, or poor decision making in critical domains.

For example we can look to Rospocher’s [2] work on building Event Knowledge graphs from News articles. The evaluation presented in the paper is insufficient because it focuses narrowly on triple accuracy and a strict assessment of event coreference resolution, neglecting many critical features necessary for a comprehensive evaluation of an EKG. It does not cover essential dimensions such as temporal reasoning and ordering, contextual completeness, fitness for use in downstream tasks, and many more.

Thus for EventPlus, [2] which relies on accurate temporal relationships, this poor evaluation fails to detect gaps in how events are ordered or connected over time. This means that when the graph is queried for complex sequences—such as determining what led to an event—it might provide misleading or incomplete answers. In practical use, this can cause errors in forecasting, decision-making, or any application built on Event-Plus that depends on precise event timelines, ultimately undermining the reliability of the entire system.

The problem is further compounded by the diverse range of EKG applications, which require different evaluation metrics and approaches. For example, evaluating an EKG for news analytics may focus on its ability to track event sequences over time and extract causal relationships, while an EKG for geographic data may prioritize spatio-temporal reasoning and completeness in representing dynamic event data. The absence of a universally accepted evaluation framework limits the comparability of different EKG models and hampers the development of more advanced and effective EKGs.

4 APPROACH

4.1 Overview

To address this gap, there is a clear need for a comprehensive and standardized evaluation framework tailored specifically for Event-Centric Knowledge Graphs (EKGs). While some existing methods focus on narrow metrics such as triple accuracy or basic temporal consistency, they often overlook essential features like event granularity, contextual completeness, and reasoning/inference capabilities. Therefore, this project will develop and validate a holistic framework that:

1. Integrates Existing Metrics (e.g., Precision, Recall, Temporal Consistency) to leverage proven evaluation methods.
2. Introduces Novel Metrics—in particular, those targeting event granularity, context-richness, causal inference, and fitness for downstream tasks.
3. Enables Iterative Testing and Refinement using benchmark datasets and comparative analyses with traditional evaluation approaches.

The remainder of this section details the multi-phase methodology—from systematically analyzing current EKG evaluation methods and identifying critical gaps, to designing and implementing the evaluation framework, and finally validating its effectiveness against established benchmarks and real-world needs.

4.2 Methods

4.2.1 Framework Design and Conceptualization

Systematic Review of Existing Evaluation Methods

1. Identify all relevant metrics currently in use for EKG evaluation—e.g., Precision/Recall for event matching, Temporal Consistency for temporal ordering, Semantic Consistency for entity alignment, and more specialized ones (e.g., coverage or integration quality).
2. Collect underrepresented metrics from the literature (e.g., event granularity, contextual completeness, causal depth) that, while mentioned, are not consistently applied in existing frameworks.

Identification of Core EKG Characteristics

1. Working with prior research, define which EKG features are critical to robust evaluation. This includes temporal sequence representation, causal link accuracy, contextual richness, reasoning/inference capabilities, and fitness for use in downstream tasks.
2. Group these features into clear evaluation dimensions—for instance, temporal accuracy, causal inference capability, semantic alignment, context depth, and more.

Initial Consolidated Framework

1. Combine all existing metrics and the underrepresented ones identified into a single, unified evaluation framework.
2. Develop a conceptual model that maps each metric to an EKG dimension (e.g., how the metric for “causal relationship precision” contributes to measuring overall causal inference quality).

4.2.2 Applying the Consolidated Framework to EKGs:

Benchmark Datasets Selection - Choose well-known EKG datasets such as EventKG, AugGKG, and ASER to test and validate the integrated framework.

Framework Application - Apply each included metric across the selected EKGs, generating detailed performance results for each dimension (temporal accuracy, contextual richness, etc.). Make sure to document any challenges (e.g., missing data, complexity in linking sub-events) that arise during implementation and scoring.

Gap Analysis - Based on the results, analyze where existing metrics and underrepresented metrics collectively still fall short—i.e., which critical EKG features remain inadequately measured.

This analysis determines the specific gaps that require novel metrics or approaches.

4.2.3 Development of New Metrics and Methods:

Identifying Uncovered Dimensions

- From the gap analysis, isolate particular areas (e.g., deep causal chain inference, advanced event granularity) where additional evaluation is needed.

Designing Novel Metrics

- Develop new or refined metrics that address the discovered gaps. For example: Advanced Causal Depth Score to evaluate multi-step causal chains beyond immediate event links and Contextual Completeness Index that quantifies how thoroughly each event is described (participants, location, time, sub-events).
- Each new metric should align seamlessly with the existing dimensions in the initial consolidated framework.

Implementation and Testing

- Incorporate these new metrics into the existing codebase or evaluation tool.
- Re-run evaluations on benchmark datasets and compare results before and after each new metric’s inclusion, confirming that the added metric provides new, actionable insights.

4.2.4 Iterative Refinement and Comparative Analysis:

1. Prototype Testing and Feedback

- Conduct iterative testing with the **updated** framework (i.e., existing metrics plus newly introduced ones).
- Collect feedback regarding **usability**, **accuracy**, and **coverage** from both technical testers and domain stakeholders.

2. Comparative Study with Existing Evaluation Methods

- Evaluate how well the refined framework addresses EKG aspects missed by traditional approaches.
- Highlight improvements, particularly in measuring multi-event causal relations, fine-grained contextual data, and the EKG’s value in real-world downstream

tasks.

3. Documentation and Dissemination

- Produce comprehensive **technical documentation** for the final framework, detailing how to apply both legacy and new metrics.
- Publicize findings through **academic outlets** and workshops, encouraging broader adoption and collaboration.

4.3 Expected Advantages of This Phased Approach

Evidence-Driven Innovation. By **first** applying a consolidated set of current and underrepresented metrics, the project ensures that any new metric is introduced only to cover actual, empirically demonstrated needs.

Reduced Redundancy This strategy avoids reinventing established metrics unnecessarily and promotes **cohesive integration** of well-known metrics with newly proposed ones.

Improved Reliability and Practical Relevance . Systematic testing on real EKGs, followed by targeted improvements, ensures that the final framework is **robust, comprehensive, and domain-**

4.3.1 Materials

Datasets: Benchmark datasets of Event-Centric Knowledge Graphs such as EventKG, AugGKG, and ASER for testing and validation purposes. Software Tools: Programming environments (Python, R), libraries for graph analysis (NetworkX, Neo4j), and machine learning (scikit-learn, PyTorch) to develop and implement the framework.

5. EVALUATION PLAN

The evaluation plan for this project aims to systematically assess the effectiveness and comprehensiveness of the proposed evaluation framework for EKG. Given the complexity and the multifaceted nature of EKGs, the evaluation will focus on validating both the framework and the newly developed metrics across a diverse range of scenarios and datasets. The following steps outline the comprehensive evaluation plan

5.1 Evaluation Objectives

The primary objectives of the evaluation are to:

Validate the effectiveness of the proposed evaluation framework in capturing the critical dimensions of EKGs, including temporal accuracy, causal inference, contextual richness, semantic consistency, and multimodal integration.

Measure Causal and Contextual Capabilities - Determine how accurately the framework evaluates direct and indirect causal relationships among events. Further , to examine the robustness of contextual integration, including location, participants, and event-specific attributes.

Validate Metrics and Their Real-World Alignment

.Ensure that new and existing metrics are meaningful, measurable, and aligned with real-world EKG outcomes. Confirm that temporal accuracy, semantic consistency, contextual richness, and other underrepresented metrics

offer actionable insights for practitioners.

Compare and Benchmark - Benchmark the proposed framework against traditional evaluation methods, highlighting advantages in capturing EKG complexities such as causal inference and event granularity. Document any relative strengths or weaknesses across different EKG models (e.g., news analytics, geographic data).

5.2 Evaluation Methodology

The following phases detail how the new evaluation framework will be systematically tested, refined, and validated to ensure it addresses the multifaceted nature of Event-Centric Knowledge Graphs . While the framework will be applied to benchmark EKGs, each phase emphasizes how the results and feedback inform an assessment of the framework's own quality—its coverage, accuracy, usability, and real-world relevance.

Phase 1:Applying Existing Metrics Within the Framework

Objective: Determine baseline effectiveness of the framework by consolidating commonly used metrics (e.g., Precision, Recall, Temporal Consistency) and underused ones (e.g., Contextual Completeness, Event Granularity) into a single evaluation suite—and then assessing how well this consolidated set performs in practice.

Procedure

1. Implementation in the New Framework

- Integrate both standard and less frequent metrics into a **prototype version** of the framework.
- Ensure each metric's role (e.g., measuring causal relationships, contextual depth) is well-documented.

2. Evaluation on Benchmark EKGs

- Run the framework on datasets such as **EventKG, AugGKG, and ASER**.
- Observe **how the framework handles** different event volumes, domain-specific structures, and data complexities.

3. Framework Performance Criteria

- **Coverage:** Do the existing metrics collectively evaluate all crucial EKG features (temporal, causal, context)?
- **Reliability:** Are metric calculations stable across different EKGs, or do certain graphs cause errors/inconsistencies?
- **Actionability:** Does the framework produce **clear** results ?

4. Outcome

- Baseline measurement of **how well the initial framework** meets its goals (completeness, clarity).
- **Evidence** of where the framework struggles (e.g., missed causal subtleties, incomplete contextual scoring), guiding the next phase.

Phase 2: Identifying and Filling Gaps With New Metrics

Objective: Evaluate how effectively adding or refining metrics (based on Phase 1 gaps) improves the framework's coverage, depth, and ability to capture key EKG nuances.

Procedure

1. **Gap Analysis:** Examine **Phase 1 results** to see which features were poorly measured or missed altogether (e.g., indirect causal links, advanced temporal overlaps).
2. **Introduction of New or Refined Metrics:** Incorporate novel metrics or revise existing definitions (e.g., an "Event Causal Depth Score" or a "Contextual Completeness Index"). Embed these into the framework's **codebase**, ensuring compatibility with existing workflows.
3. **Re-Evaluation Re-run** the upgraded framework on the same EKG datasets to compare "**pre-gap**" vs. "**post-gap**" results. Evaluate whether the new metrics capture previously unrecognized aspects of EKG performance and provide **more actionable** insights for EKG developers or domain practitioners.
4. **Framework Performance Criteria** . This will consist of the following, **Incremental Value:** Does each new metric **add** clear diagnostic power without duplicating existing ones? And **Usability:** Are the new metrics intuitive for end-users, or do they require complex additional data?
5. **Outcome**
 - A **quantitative and qualitative** measure of how new metrics elevate the framework's completeness and real-world usefulness.
 - Concrete data on which improvements succeeded (or failed), informing further iteration.

Phase 3: Iterative Refinement of the Framework

Objective **Continuously evaluate** and improve the framework's accuracy, scalability, and ease of use, ensuring it evolves into a **robust** and **domain-ready** tool.

Procedure

1. **Stakeholder Feedback Loops**
 - Present iterative results to Maria and Zola, verifying that metrics align with **real-world** needs (e.g., in healthcare, finance, or news analytics).
 - Gather feedback on **interface design**, **result interpretability**, and whether the framework's outputs guide better decisions.
2. **Performance and Reliability Checks**
 - Conduct repeated evaluations with the newly refined metrics to see if issues like **algorithmic bottlenecks** or data-handling errors persist.
 - Document any changes that **increase** the framework's reliability or reduce computational overhead.
3. **Criteria for Framework Readiness**
 - **Consistency:** The framework consistently produces stable, replicable scores across

repeated runs and different EKG datasets.

- **Scalability:** The framework handles larger or more complex graphs without major performance drops.
- **User Acceptance:** Practitioners indicate that results are clear, relevant, and **inform** real EKG improvements.

4. Outcome

- A refined framework whose iterative updates are **tracked and validated**, with each iteration bringing measurable benefits (e.g., expanded coverage, greater efficiency).

Phase 4: Comparative Study With Existing Evaluation Methods

Objective **Formally evaluate** whether the final version of the framework outperforms or complements **traditional EKG evaluation** techniques, thereby validating the entire iterative process.

Procedure

1. **Side-by-Side Testing**
 - Select one or more **established** evaluation methods (e.g., triple-accuracy-based approaches) as benchmarks.
 - Run both the **final** iteration of the new framework and the **existing** methods on a representative set of EKGs (possibly including domain-specific ones).
2. **Criteria for Comparison**
 - **Depth of Coverage:** Which method better captures advanced causal chains or extensive contextual data?
 - **Practical Usability:** How do experts perceive the ease of setup, result clarity, or overall utility for real-world tasks?
 - **Quantitative Scores:** Compare final metrics, e.g., is the new framework's contextual completeness score significantly better in reflecting real relationships?
3. **Outcome**
 - **Demonstration** that the new framework either (a) provides **unique insights** or (b) complements older methods with a broader or deeper view of EKG performance.
 - A conclusive set of **best practices** for EKG evaluation, highlighting the strengths and limitations discovered.

Summary of Framework Evaluation

Through these **four phases**, the **framework itself** is consistently under scrutiny—first tested with existing metrics, then enhanced with new metrics, continuously refined based on real feedback, and finally **benchmark-compared** to validate its overall superiority or complementarity. Each step:

1. **Generates evidence** about how well the framework meets its goals (comprehensiveness, usability).
2. **Pinpoints and addresses** shortcomings, ensuring a continuous improvement loop.

3. **Demonstrates** practical value through direct comparisons with established methods and real-world feedback.

By the end of Phase 4, the project will have a clear **body of data** demonstrating **how** the new framework evolved, **why** certain metrics were introduced, and **how effectively** it addresses the complexities of event-centric knowledge representation.

6.EXPECTED OUTCOMES

The project aims to achieve the following key outcomes:

A comprehensive Evaluation Framework for Event Centric Knowledge Graphs:

A new, standardized evaluation framework that integrates existing evaluation metrics and introduces novel ones specifically tailored to assess the unique characteristics of EKGs. The framework will provide a holistic assessment that covers critical dimensions such as temporal accuracy, causal inference, contextual richness, semantic consistency, and multimodal integration.

Metrics for Evaluating EKGs:

Introduction of specific metrics, ensuring that EKGs are not only theoretically robust but also practically valuable for applications like event prediction, risk assessment, and recommendation systems.

A Functional Tool for EKG Evaluation:

Development of a user-friendly software tool that implements the new evaluation framework. This tool will enable researchers and practitioners to input their EKGs, select relevant metrics, and generate comprehensive evaluation reports, facilitating the use of the framework in both research and practical applications.

Improved Understanding of EKG Strengths and

Limitations: The evaluation framework will enable a balanced understanding of the strengths and limitations of various EKGs by capturing the interconnected and multifaceted nature of their features. This will help in identifying areas for improvement and guide the development of more advanced and effective EKGs.

Enhanced Comparability of EKG Models:

The standardized evaluation framework will allow for more meaningful comparisons between different EKG models. This comparability is essential for advancing the field of event-centric knowledge representation and fostering innovation.

Validation of the Framework Through Comparative

Studies: Rigorous validation and comparative studies will be conducted to demonstrate the effectiveness of the new evaluation framework against existing methods. This will provide empirical evidence of the framework's robustness and reliability.

Contributions to Advancing Knowledge Representation

in AI: By providing a more robust evaluation framework for EKGs, the project will contribute significantly to the fields of artificial intelligence, knowledge engineering, and data science, contributing to better decision-making and

analytical capabilities in real-world scenarios. These expected outcomes will collectively ensure that the project delivers a comprehensive solution to the current challenges in evaluating Event Centric Knowledge Graphs, setting new standards for research and practical applications in this domain.

6. 7. POTENTIAL CHALLENGES

6.1 Development of Novel Metrics

The Challenge: Creating new metrics involves significant theoretical and practical challenges. Defining these metrics to be meaningful, measurable, and universally applicable across different EKGs and domains is difficult.

Impact: Without properly defined and validated metrics, the framework may not accurately assess the "Fitness for Use in Downstream Tasks," compromising its credibility and effectiveness.

How to Address It: Employ an iterative development approach, allowing for refinement based on pilot testing and feedback. Document and publish preliminary findings to engage the research community and gather broader input.

6.2 Data Availability and Quality

The Challenge: Accessing high-quality benchmark datasets like EventKG, AugGKG, and ASER may be restricted due to licensing issues or incomplete coverage of necessary features, limiting testing and validation opportunities.

Impact: Limited or low-quality data can hinder comprehensive testing, affecting the robustness and generalizability of the evaluation framework.

How to Address It: Seek publicly available datasets or establish partnerships with organizations for data access. Utilize data augmentation techniques or generate synthetic datasets to simulate required features. Implement data preprocessing to enhance quality where possible.

6.3 Comparative Analysis Difficulties

The Challenge: Conducting fair and meaningful comparisons with existing evaluation methods is complex due to differences in evaluation criteria, metrics, and assumptions.

Impact: Difficulty in comparative analysis may affect the credibility of validation efforts, making it challenging to demonstrate the new framework's advantages.

How to Address It: Align evaluation criteria with existing standards where possible. Clearly document methodologies and justify any deviations. Use standardized datasets and controlled experiments to facilitate fair comparisons.

6.4 Computational Resource Constraints

The Challenge: Processing and evaluating large-scale EKGs demand significant computational resources. Limited access to high-performance computing may slow down experiments and testing phases.

Impact: Insufficient computational resources could lead to project delays and limit the scope of testing, affecting the comprehensiveness of the research.

How to Address It: Secure necessary computational resources early, possibly through institutional support or cloud computing services. Optimize algorithms for efficiency and plan experiments strategically to manage resource usage effectively.

9. CONCLUSION

This project proposes a comprehensive evaluation framework for Event Centric Knowledge Graphs (EKGs) to address the limitations of current fragmented methods that fail to capture the full complexity of EKGs, including temporal, causal, and contextual relationships. Existing evaluations often overlook these dimensions, resulting in incomplete assessments that hamper EKGs' effectiveness in real-world applications.

The proposed framework integrates existing evaluation methods and introduces new metrics, such as Task-Specific Performance Score (TSPS) and Task Adaptability Rate (TAR), to provide a more holistic and meaningful evaluation of EKGs. By developing a user-friendly tool and validating it with benchmark datasets, the framework will allow for better comparison of EKG models and guide future improvements. This project aims to set a new standard for evaluating EKGs, ensuring they are robust, versatile, and fully capable of supporting advanced analytics and decision making across diverse domains. The outcomes will significantly advance research and practical applications in knowledge representation.

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Project Plan:

Month	Key Activities	Deliverables
February 2025	- Metric Consolidation & Baseline Setup • Finalize existing and underused metrics (Precision, Recall, Temporal Consistency, Contextual Completeness). • Integrate them into the baseline framework (Python/Neo4j/NetworkX). • Prepare benchmark datasets (EventKG, AugGKG, ASER). - Initial Results & Documentation • Run the baseline framework on datasets to get preliminary scores. • Note coverage gaps and data-handling issues.	- Consolidated Baseline Framework with integrated metrics. - Baseline Results Report summarizing performance and key shortcomings.
March 2025	- Gap Analysis & Metric Dev Planning • Review February results to identify under-measured features (e.g., multi-step causal chains). • Get feedback from domain experts to confirm real-world importance. - Metric Enhancement Strategy • Draft plan for new/refined metrics (e.g., “Causal Depth Score”). • Outline required data structures/algorithms. - Thesis: Literature Review • Incorporate baseline findings to refine the literature review.	- Gap Analysis Document detailing which EKG aspects need better metrics. - Enhanced Metric Design Plan with proposed formulas and data requirements. - Initial Literature Review chapter draft.
April 2025	- Develop & Integrate New Metrics • Implement new metrics targeting uncovered dimensions (e.g., advanced causal inference). • Update framework code and documentation accordingly. - Limited Scope Testing • Run tests on small dataset samples to verify correctness and performance overhead.	- Extended Framework featuring newly introduced/refined metrics. - Documentation describing metric calculations and how they address identified gaps.
May 2025	Re-Evaluation & Iterative Refinement • Re-run the updated framework (with new metrics) on all benchmark datasets. • Compare “pre-gap” vs. “post-gap” performance to quantify improvements. - Metric Tuning • Tweak definitions if unexpected results arise. - Thesis: Methodology Draft • Begin writing methodology chapter explaining metric integration and validation approach.	- Comparison Report showing improvements vs. baseline framework. - Early draft of the Methodology chapter.
June 2025	- Stakeholder Feedback & Optimization • Present results and metrics to experts (e.g., healthcare, finance, GIS) to ensure real-world relevance. • Gather feedback on usability, clarity, and interpretability. - Framework Optimization • Address performance bottlenecks or data-handling inefficiencies. - Thesis: Results Chapter (Initial) • Start drafting sections with tables/figures summarizing improved evaluation outcomes.	- Refined & Optimized framework incorporating expert suggestions. - Initial Results chapter draft with charts and case studies.
July 2025	- Early Comparative Analysis • Benchmark the refined framework against at least one traditional EKG evaluation method. • Note differences in coverage, ease of use, computational performance. - Further Refinement • Adjust code or metrics based on comparative insights. - Update Documentation • Keep methodology/results sections current if significant changes occur.	- Interim Comparative Analysis Report detailing advantages over older methods. - Updated methodology/results documentation if additional refinements are made.
August 2025	- Comprehensive Testing & Documentation • Conduct scalability tests on larger or more complex EKGs to confirm framework stability. • Record final optimizations or bug fixes. - Thesis Writing • Refine and polish the methodology and results chapters, integrating final test findings. - User-Facing Docs • Prepare tutorials or instructions for broader adoption of the framework.	- Complete & Stable Framework ready for widespread application. - Polished drafts of the Methodology and Results chapters.

September 2025	- Final Comparative Study & Thesis Completion • Perform a full, side-by-side comparison of the final framework vs. traditional methods across multiple EKG scenarios. • Gather both metric-based and user-based feedback. - Discussion & Conclusion • Write concluding sections on the framework's strengths, limitations, and future directions. • Prepare the entire thesis for submission, formatting per institution guidelines.	- Final Comparative Study Report detailing the framework's performance across diverse EKGs. - Complete Draft Thesis (all chapters near final form).
October 2025	- Final Revisions & Thesis Submission • Address any last-minute supervisor or committee feedback. • Prepare presentation/defense materials (if required). - Official Submission • Submit the master's thesis. • Release final framework (code + documentation) to the public or departmental repository.	- Master's Thesis Submission by the end of October. - Framework Public Release with documentation and source code.

Feature to Evaluate	Why	Metrics
Accuracy of Event Representation	Ensures that the EKG accurately reflects the events, including their attributes (e.g., time, location) and relationships. Crucial for event prediction, decision-making, and knowledge discovery.	Precision, Recall, F1-Score, Entity Resolution Accuracy, Relation Extraction Accuracy
Temporal Reasoning and Ordering	EKGs need to handle temporal relations and sequences effectively, understanding event duration, intervals, and their order of occurrence.	Temporal Consistency, Event Sequencing Accuracy, Temporal Query Precision
Contextual Completeness	Evaluates if the EKG captures necessary context around events, such as participants, causes, effects, and concurrent events, essential for generating meaningful insights.	Coverage of Contextual Attributes, Richness of Contextual Information
Scalability and Performance	Measures the ability of the EKG to handle large-scale data and complex queries without performance degradation, crucial for real-time applications and big data environments.	Query Response Time, Scalability Benchmarking, Indexing Efficiency
Interoperability and Integration	EKGs should easily integrate with other knowledge graphs, databases, and systems, enabling interoperability for broader applications.	Compliance with Ontology Standards (e.g., OWL, RDF), Ease of Data Integration, API Compatibility
Reasoning and Inference Capabilities	Evaluates the EKG's ability to support logical reasoning and infer new knowledge from existing data, important for automated decision-making processes.	Deductive Reasoning Accuracy, Inductive Reasoning Accuracy, Consistency of Inferred Data
Flexibility and Extensibility	Measures how easily new data, entities, relationships, and event types can be added or modified in the EKG, critical for long-term usability and adaptability.	Schema Modification Ease, Ontology Extension Capability, Support for Dynamic Schema Evolution
Trustworthiness and Provenance	Ensures the reliability of the data in the EKG by maintaining detailed provenance information about the sources of events and any transformations applied.	Provenance Coverage, Data Trustworthiness Score, Source Transparency
Support for Multimodal Data (Where applicable)	Assesses the EKG's ability to handle different types of data (e.g., text, images, audio) that provide richer event representations.	Multimodal Integration Quality, Cross-Modal Query Performance
Semantic Richness and Ambiguity Handling	EKGs must capture the semantic nuances of events and handle ambiguous or conflicting information effectively.	Ambiguity Resolution Rate, Semantic Consistency Score, Polysemy Handling Capability
Causal Relationship Representation and Reasoning	Understanding causal relationships between events is fundamental for predicting outcomes, understanding dependencies, and making informed decisions based on potential cause-and-effect chains.	Causal Precision and Recall, Causal Inference Accuracy, Causal Chain Consistency, Counterfactual Reasoning Capability, Temporal Causality Accuracy, Causal Graph Completeness, Causality Strength and Directionality
Event Granularity and Hierarchical Representation	EKGs should be able to represent events at different levels of granularity and show hierarchical relationships between events.	Granularity Flexibility Score, Hierarchical Consistency, Sub-event Linking Accuracy
Cross-lingual and Multilingual Support	In a globalized world, EKGs need to handle events described in multiple languages or translate event information across languages.	Cross-lingual Consistency, Multilingual Query Performance, Translation Accuracy
Event Coreference Resolution	Ability to identify when different descriptions or mentions refer to the same event, crucial for maintaining consistency and avoiding redundancy.	Event Coreference Accuracy, Cross-source Event Linking Precision
Spatio-temporal Reasoning	Events often have complex spatio-temporal relationships that go beyond simple temporal ordering.	Spatio-temporal Query Accuracy, Geographical Reasoning Capability
Dynamic Event Updating and Versioning	As new information becomes available, the EKG should be able to update event representations and maintain version history.	Update Propagation Speed, Version History Completeness, Temporal Consistency of Updates
Event Importance and Relevance Scoring	Not all events are equally important; the EKG should be able to assess and represent the significance of different events.	Relevance Ranking Accuracy, Important Event Detection Rate

Table 2: Key Milestones and Deliverable

Milestone Title	Description	Deliverables	Timeline
Literature Review and Criteria Identification	Conduct a comprehensive review of existing evaluation methods for EKGs and related knowledge graphs. Identify gaps and necessary evaluation criteria for the new framework. Engage with domain experts to refine these criteria further.	- Detailed literature review report - List of identified gaps in current methods - Preliminary table of evaluation criteria and metrics	Done
Definition of Evaluation Dimensions	Define core evaluation dimensions such as temporal accuracy, causal inference capability, and contextual richness. Develop a conceptual model representing these dimensions and their interrelationships to guide the development of evaluation metrics.	- Document outlining defined evaluation dimensions and sub-components - Conceptual model for the evaluation framework	September 2024
Development of Novel Evaluation Metrics	Develop new evaluation metrics based on identified gaps to comprehensively assess EKGs. Integrate these metrics with traditional ones to address aspects like temporal reasoning, causal relationships, and contextual information. Introduce metrics like Task-Specific Performance Score (TSPS) and Task Adaptability Rate (TAR) to evaluate the utility of EKGs for downstream tasks.	- Set of newly developed evaluation metrics - Detailed documentation for application and interpretation	December 2024
Development of Evaluation Framework and Tool	Develop algorithms and models to implement the proposed evaluation metrics and integrate them into a cohesive framework. Create a software tool with a user interface to facilitate practical application, allowing users to apply the framework to their EKGs and generate comprehensive evaluation reports.	- Prototype evaluation tool with user interface - Source code for algorithms and models - Comprehensive user guide and technical documentation	January 2025
Iterative Refinement of Framework	Refine the evaluation metrics and framework based on feedback from testing. Adjust algorithms, modify metric definitions, and enhance the	- Revised framework and metrics - Updated documentation	April 2025

	modularity of the framework to accommodate different types of EKGs.	reflecting changes	
Comparative Study and Validation	Conduct a comparative analysis of the proposed framework against existing evaluation methods to validate its effectiveness. Apply both the new framework and traditional methods to various EKG models to assess their strengths and weaknesses.	- Comparative study report - Validation report detailing strengths and areas for improvement of the framework	May 2025
Documentation and Dissemination	Develop comprehensive documentation for the framework, including a user guide, technical specifications, and guidelines for interpreting results. Disseminate findings through publications in academic journals, conferences, and workshops to promote adoption and further development of the framework by the research community.	- Complete documentation package - Published research papers - Participation in conferences and workshops	June 2025